Divyansh Joshi

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Professional Summary

Systems & Distributed Software Engineer specializing in low-latency infrastructure, distributed consensus algorithms (Raft), concurrent network programming, and AI-augmented systems engineering. Extensive experience building production services in Rust (Tokio, Leptos, WASM), Python, and C/C++, coupled with Model Context Protocol (MCP), multi-agent AI orchestration, Linux kernel internals, and ZFS storage pipelines.

Professional Experience

LNJ Corporate Solutions Indore, India
Systems & Infrastructure Engineer Dec 2025 - Present

- **Distributed Storage & Virtualization:** Designed and maintained a multi-tier storage cluster on Proxmox VE utilizing TrueNAS SCALE and ZFS RAID-Z arrays, enforcing automated block-level snapshot replication with 99.99% reliability.
- **High-Performance Search Engine:** Deployed an optimized multithreaded indexing engine in Python over 350+ GB (700,000+ files), reducing search query latencies by 85% with near-zero host CPU footprint.
- **Zero-Trust Network Protocol Mesh:** Spearheaded encrypted transport overlays using Tailscale (WireGuard) and Cloudflare Tunnels (mTLS) to secure internal RPCs and administrative daemons from WAN exposure.
- **AI-Assisted Systems Telemetry:** Automated asynchronous service daemons interfacing with self-hosted LLM endpoints and TrueNAS REST APIs to analyze real-time I/O performance and SMART storage diagnostics.

Projects

PeerStor | Fault-Tolerant Distributed Storage Cluster (*Rust, Tokio, Raft, gRPC*) 2025 - 2026

- Architected an asynchronous peer-to-peer storage engine across 10+ virtualized nodes, implementing Raft-backed leader election to guarantee deterministic data recovery from node failover in <300ms.
- Built concurrent chunking and stream replication pipelines in Rust, achieving sub-20ms synchronization latency across clustered nodes with zero data corruption under simulated partition failures.

Bare-Metal IoT Gateway | Remote Access & Power Governor (*ESP32 C++, LittleFS, HTTPS*) 2025

- Formulated an embedded gateway on ESP32 microcontrollers executing encrypted Wake-on-LAN (WoL) and ACPI power-state transitions over authenticated HTTPS, securing out-of-band management without WAN exposure.
- Integrated TrueNAS REST API clients with persistent cron scheduling on non-volatile LittleFS flash memory, guaranteeing 100% deterministic task recovery across 50+ simulated power outages.

WASM Edge Platform | Medical AI Decision Engine (*Rust, Leptos, WASM, Cloudflare, LangGraph*) 2026

- Engineered a client-side WebAssembly platform with sub-50ms Time-to-Interactive (TTI) and a 100/100 Lighthouse score utilizing Rust and Leptos, deployed over Cloudflare edge runtime for globally distributed zero-cold-start performance.
- Constructed an autonomous LangGraph decision pipeline extracting 10,000+ unstructured medical records into validated biomarker JSON datasets, querying clinical trial APIs with sub-3-second inference speeds.

Technical Skills

Languages: Rust (Tokio, Async-std, Leptos, WebAssembly), Python (FastAPI, Asyncio, Pandas), C/C++ (Embedded/ESP32), SQL, Bash Scripting.

Distributed Systems & Protocols: Raft Consensus, gRPC, Protocol Buffers, Peer-to-Peer Networking, Zero-Trust Architecture (mTLS), High Availability, ZFS Storage.

AI Engineering & Tooling: Model Context Protocol (MCP), Multi-Agent AI Orchestration, LangGraph, LangChain, Self-Hosted LLMs (Groq, Ollama), Context Architecture.

Infrastructure & DevOps: Linux Kernel Tuning, Proxmox VE, TrueNAS SCALE, Docker, Kubernetes, Tailscale (WireGuard), Cloudflare Workers, AWS, GCP, CI/CD (GitHub Actions), Prometheus.

Certifications & Accreditations

Cloud & Database: Oracle Cloud (OCI) Foundations Associate | MongoDB Node.js Developer Path

Systems Automation & AI: HashiCorp Certified: Terraform Associate (003) (In Prep) | DeepLearning.AI – LangChain Development

Education

Vellore Institute of Technology (VIT) Bhopal, India
Bachelor of Technology in Computer Science and Engineering 2022 - 2026